

SE 413/513

Software Testing

Fall 2026 Syllabus

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Meeting Times: TR 4:00 pm–5:15 pm
(DCIH 505)
Office Hours: Online / MAK D-2-216
([schedule by appointment](#))

1. General Information

1.1 Course Description

This course introduces students to fundamental software testing principles and their practical application using Python-based testing tools. Students will learn to design, implement, automate, execute, and evaluate tests across multiple levels of the software development lifecycle. The course integrates contemporary practices in automated testing, continuous integration, API and web testing, and selected advanced testing techniques while preserving the core topics of the existing course description. Major topics include *basic testing terminology and concepts; unit testing and unit testing tools; control-flow and data-flow testing; coverage-based and functional testing; integration and system testing, including test design, planning, and execution; acceptance testing; and software technical reviews.*

1.2 Course Expectations and Learning Outcomes

Students completing SE 413/513 should be able to:

- Explain fundamental software testing terminology, principles, processes, and the role of testing throughout the software development lifecycle, including distinctions among unit, integration, system, and acceptance testing.
- Design and implement systematic test cases using black-box, white-box, control-flow, data-flow, and coverage-based testing techniques, and develop automated unit tests in Python using pytest.
- Execute and evaluate unit, functional, integration, system, and acceptance tests, including the development of appropriate test plans and scenarios, interpretation of test results, and measurement and interpretation of code coverage.
- Apply and evaluate software testing tools and practices within a software-development project, including version control and continuous integration to support automated testing, while recognizing the strengths and limitations of contemporary testing techniques and tools.
- Conduct technical reviews and communicate testing outcomes by documenting and reporting defects, risks, findings, recommendations, and relevant testing activities clearly and effectively.

1.3 Requirements and Prerequisite

Students need an internet connection, a computer, access to Blackboard, and the ability to run Python and install required Python packages. If you have difficulty accessing these resources or meeting these requirements, contact the instructor so that an appropriate solution can be explored.

Prerequisite:

- SE 413: [CIS 350 \(Introduction to Software Engineering\)](#)
- SE 513: Admission to MS in applied computer science, artificial intelligence, or software engineering.

1.4 Delivery Format

The course is delivered in multiple delivery formats, supported by Blackboard for course materials, assignments, announcements, and grades.

1.5 Textbooks and Reading References

Specific references will be shared throughout the course. The following resources are recommended:

- Paul Ammann and Jeff Offutt, *Introduction to Software Testing*, 2nd ed., Cambridge University Press, 2016.
- Mauricio Aniche, *Effective Software Testing: A Developer's Guide*, Manning, 2022.
- [ISTQB Certified Tester Foundation Level \(CTFL\) v4.0.1 Syllabus](#).

2. Tentative Schedule

Subject to change.

Week no.	Week of	Topic
1	Aug. 31	Testing Foundations
2	Sept. 7	Unit Testing
3	Sept. 14	Test Case Design
4	Sept. 21	TDD and Test Doubles
5	Sept. 28	Control-Flow Testing
6	Oct. 5	Data-Flow Testing and Coverage Concepts
7	Oct. 12	Coverage-Based Testing
8	Oct. 19	Functional Testing
9	Oct. 26	Integration Testing
10	Nov. 2	System Testing
11	Nov. 9	Acceptance Testing
12	Nov. 16	Project / Testing Workshop
13	Nov. 23	Continuous Testing
14	Nov. 30	Advanced Testing, Emerging Techniques, and AI-Assisted Testing
15	Dec. 7	Project Presentation

Week no.	Week of	Topic
16	Dec. 14	Exam & Final Evaluation Week

Final Examination Period

The SE 413/513 final examination will be administered during the University examination period. Class instruction ends December 12, 2026, and the examination period is December 12–19, 2026. The official final-exam date, time, and location will follow the University final-examination schedule and will be communicated through Blackboard.

3. Course Organization and Grading Format

3.1 Grading Distribution

The final course grade will be determined using the following grading distribution:

Assessment	Weight
Testing exercises / programming assignments	30%
Exams / quizzes	40%
Team software-testing project	20%
Final presentation / advanced analysis	10%
Total	100%

3.2 Exam and Quiz Proctoring Requirements

To help maintain the integrity of the course, all exams—including quizzes—will be conducted using the **Respondus LockDown Browser** with the webcam enabled. Detailed instructions will be shared before each exam or quiz to guide you through the process.

Please take a moment to ensure that the LockDown Browser is installed and set up ahead of time. You can find helpful instructions in the guide below. Feel free to reach out to me if you have any questions or need assistance—I am happy to help.

[Respondus LockDown Browser Installation and Setup Guide](#)

3.3 Semester Project

Students work in small teams on a realistic Python software system. Each group will identify an existing or newly developed software system and conduct an end-to-end testing project. The project emphasizes a learning-by-doing approach in which students apply software testing principles, test-case design techniques, automated testing, and appropriate testing tools to evaluate the quality and reliability of the selected software system.

Teams will develop and execute appropriate tests, document test results and defects, and communicate their findings and recommendations. Depending on the project, testing activities may include unit, functional, integration, system, and acceptance testing, as well as automated regression and coverage analysis.

Each team will deliver a final presentation and submit project documentation. All members must

explicitly describe their individual and collective contributions to the project.

3.4 Grading Scale

For final grade calculation, any fractional final percentage will be rounded up to the next whole percentage. For example, 89.1% will be recorded as 90%.

Grade	Minimum	Grade	Minimum
A	93%	C	73%
A-	90%	C-	70%
B+	87%	D+	67%
B	83%	D	60%
B-	80%	F	Below 60%
C+	77%		

4. Course Expectations, Policies, and Academic Honesty

4.1 Expectations

To support your success in this course, you are expected to:

- Check Blackboard regularly for announcements, course materials, assignments, and grades.
- Stay up to date with required course videos and readings.
- Communicate feedback about the course and teaching that may help improve the learning experience.
- Adhere to CIS and GVSU policies on academic honesty.

4.2 Course Policies

- **Individual work:** Homework and other individual assignments must be completed independently unless otherwise stated.
- **Due dates:** Specific due dates will be posted with each assignment in Blackboard. Unless otherwise announced, assignments are due at 11:59 p.m. Eastern Time on the stated due date.
- **Late policy:** 10% of the maximum possible grade is deducted per day late, up to five days (50% total). After five days, the assignment will not be accepted.
- The instructor reserves the right to modify course policies, the course calendar, and due dates as needed.
- This course is subject to the GVSU course policies listed at www.gvsu.edu/coursepolicies/.

4.3 Use of Generative AI Tools

Generative AI tools may be used only as permitted for individual course activities and assignments. Students remain responsible for understanding, testing, validating, and being able to explain all submitted work. AI-generated content or code must not be submitted as the student's own work

when prohibited by the assignment instructions. When AI assistance is used in a permitted manner, students may be required to disclose the tool used and the nature of the assistance. Specific AI-use expectations will be provided with individual assignments and assessments.

4.4 Academic Honesty

All students are expected to adhere to the academic honesty standards of Grand Valley State University and the academic honesty guidelines of the School of Computing. Details are available at <https://www.gvsu.edu/computing/academic-honesty-30.htm>.

Collaboration and learning from peers and the broader community are encouraged, but presenting another person's work as your own is prohibited. For individual assignments:

- Document collaboration.
- Do not electronically transfer code between students.
- Cite code found on the internet with an active link to the source. External code must not solve the entirety of an assigned problem or project.
- You may participate in online forums, but do not post solutions or solicit others to complete your work.
- You may discuss problems with peers in non-technical terms, but you remain responsible for your own submission.
- You are ultimately responsible for all aspects of your submissions and should be able to explain and defend your work.

5. Useful Resources

5.1 Blackboard

Course materials, assignments, grades, and announcements will be posted to Blackboard (<https://lms.gvsu.edu/>). Students are responsible for staying informed. Blackboard announcements are used to communicate timely information, including changes to deadlines; students should ensure their notification settings allow them to receive updates.

5.2 Other Academic Resources

GVSU provides academic-support resources, including:

- The Writing Center
- Speech Lab
- Research Consultants
- Library Liaisons

5.3 Disability Support

Students who need accommodations should present a memo from Disability Support Resources (DSR) describing the approved reasonable accommodations. If you have not already done so, contact Disability Support Resources at 215 CON, by phone at 331-2490, or by email at dsrgvsu@gvsu.edu. The instructor cannot provide disability accommodations until a DSR-issued memo has been received. All discussions will remain confidential. More information is available at <https://www.gvsu.edu/dsr/>.

5.4 Religious Observance

Many University students, staff, and faculty observe religious traditions. The Religious Inclusion Policy acknowledges the right of students, staff, and faculty to engage in religious observances. If you need special accommodations to observe a religious holiday, please make arrangements with the instructor in advance.

5.5 In Case of Emergency

Fire: Immediately proceed to the nearest exit when a fire alarm sounds. Do not use elevators. Additional information is available on the University Emergency website at gvsu.edu/emergency.